

Algebra 1

Unit 13

Lesson 2 Homework

Name Answer Key

Date _____

Period _____

Lakita deposited \$15,000 in a savings account with a 3.75% APR compounded monthly.

1. Write an equation that models the amount of money Lakita has in the account.

$$A = P \left(1 + \frac{r}{n} \right)^{nt} \quad A = 15,000 \left(1 + \frac{.0375}{12}^{12t} \right)$$

2. What is the balance in the account after 10 years?

$$15,000(1.003125)^{120}$$

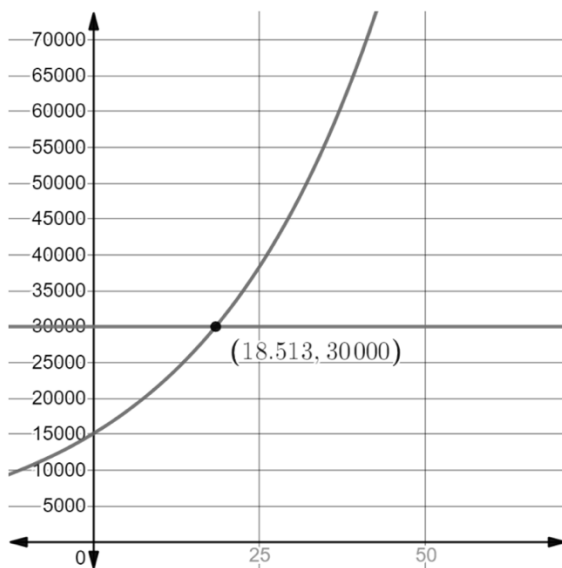
\$21,812.11

3. What is the balance in the account after 15 years?

$$15,000(1.003125)^{180}$$

\$26,302.74

4. Use the graph to determine when the savings account will reach \$30,000.



Approximately 18.5 years.

For questions 5-10, complete the table below.

	Simple Interest	Compound Interest
Investment amount and APR	\$100,000 is invested at an interest rate of 2.25% APR	\$100,000 is invested at an interest rate of 2.25% APR compounded monthly
Write the equation.	5. $A = P(1 + rt)$ $A = 100,000(1 + 0.0225t)$	6. $A = P \left(1 + \frac{r}{n}\right)^{nt}$ $A = 100,000 \left(1 + \frac{.0225}{12}\right)^{12t}$
Determine the amount after 5 years.	7. $100,000(1 + .1125)$ $\boxed{\$111,250}$	8. $100,000(1.001875)^{60}$ $\boxed{\$111,895.44}$
Determine the amount after 25 years.	9. $100,000(1 + 0.5625)$ $\boxed{\$156,250}$	10. $100,000(1.001875)^{300}$ $\boxed{\$175,413.05}$